VU AI HUB · LECTURE NOTES

Introduction to AI

Year 1, September to October 2025

Written by Inaaya and shared on Notion. Original page: cheddar-wall-f3b.notion.site/2609801a9309809e958ecb1bcf2d73c0

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1st of September (Introduction)

Problem-driven Artificial Intelligence

4 steps of research:

- Identify the problem
- Understand the context (theory)
- Model your solution
- Realize (application)

5th of September (Methodology)

Finding Literature

8th of September (Guest)

Machine Learning

"Machine learning is to automatically identify patterns from data"

"A computer program is said to learn from experience E with respect to some class of tasks T and performance P, if its performance at tasks in T improves with E"

experiences in this case are examples, machine learning gets better when the amount of examples, experiences, increases

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What are machine learning tasks?

- **Supervised learning** is the machine learning task of inferring a function from a set of labeled training data
  - Ex. On this picture this person is wearing glasses, in this picture the person is not

- in **unsupervised learning**, there is no target measure (or label), and the goal is to describe the associations and patterns among the attributes (recognizing patterns)
  - Ex. Does the machine see a particular resemblance in pictures
- **Reinforcement learning** tries to find optimal actions in a given situation so as to maximize a numerical reward that does not immediately come with the action but later in time
  - Ex. Robot driving in a maze. If the robot reaches the end it doesn't know if the turn to the right was the best idea
  - So reinforcement learning tries to find the best optimal action to get out of said maze

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Supervised learning approaches learning using a labeled dataset

- "regular" machine learning algorithms
- pattern recognition

Reinforcement learning has a few ingredients

- state, action and reward
- can also be used for personalization (Netflix recommendations, medical treatments etc.)

Steps to come to a conclusion from a machine learning algorithm

- Understanding of domain: if you want to apply machine learning you first need to figure out in what domain to apply it in
- Understanding of data: what are the different fields of the data
- Data preparation: missing values etc. to have good quality data
  - This takes 60-80% of the overall time

Evaluation

- Want to have models that predict well for new examples
- We train on a training set and apply our algorithms to an independent test set to assess generalizability
- ??

ImageNet

- Identifying to which class an image belongs
-

Deep learning

- Neural networks (deep learning algorithms) - composed of thousands of very simple computational units
- The input layer is where the deep learning model ingests the data for processing
- Connections have a strength (also called a weight)
  - the thinner the arrow, the less the importance and vice versa
- Neurons multiply the weight with the value of the input (w1)
  - Then you combine all the different multiplications by adding them up
  - Then we apply a certain function ex. if the w0 is more than … a person is wearing glasses
- How do we learn these weights?
  - The real knowledge of the network is in the weight
  - We use our network to predict what picture we are seeing

    - If we make a mistake we “punish” the weights that contributed most to the mistake, and adjust them to avoid the same mistake again
    - We continue this for our entire training set
    - Problem: there are way too many weights, way too much data to train
- Deep learning: reduces the number of weights through special layers

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AlphaGo

- It uses reinforcement learning
- We learn to select the best action in a given state such that we maximize all the future reward
- In AlphaGo: we want to win the game
- Actions in a state might give us an immediate reward (ex. we gain advantage on the board) but might turn out bad in the long run (we lose)
    - Sequential actions

Of course we have lots of states, so how do we learn what to do when?

- We just start selecting random actions to start with (exploration) and start learning which actions are suitable in what states (i.e. give us a lot of future reward)
- We move more to exploitation once we have found a good strategy, using the action that gives the biggest amount of reward which we concluded during exploration
- We mostly do this in a simulator
- We deal with the huge number of states with *deep reinforcement learning*
- We need lots of computational researches and computing power

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Generative AI

- We generate things we haven’t seen before
- Generative Adversarial Networks, they work with two competing forces at work:
    - One network tries to generate images as good as it can (generator)
    - The other network tries to figure out which images are real and which ones are fake (discriminator)
        - If the other network figured it out, it wasn’t good enough, but if it didn’t figure it out is was done well
    - So we need real data for this to work

ChatGPT

- Uses a combination of learning techniques (and a “safety net”)
- Driven by a language model (input and output are both language)

12th of September (Guest)

Knowledge Representation

15th of September (Methodology)

Basics of Computers

Book: Artificial Intelligence, A Modern Approach - Stuart Russell & Peter Norvig

Computers - any system device that manipulates (= stores, retrieves, processes) data

Computer Architecture

- User
- Application (software)
- Operation system (software)
- Hardware (physical)

Hardware

1. Peripheral devices
   - Microphone
2. Output devices
   - Monitor, speaker, printer etc.
3. Input devices
   - Keyboard, mouse, etc.
4. Internal components
   - CPU, RAM (= quick access memory) (HDD, CD), SSD (faster than the hard disk)

Processors/CPU's

- calculations and operations on the data
- execute instructions
- Arithmetic and Logical Unit ALU: (addition, subtraction, multiplication, division and also making logical decisions like > or <, etc)
- Control Unit (CU): execution of the tasks

Memory/RAM

- Main, primary memory of the computer
- Stores data and instructions, while the processing is taking place in the CPU
- volatile (data is lost when the power is disconnected)

Another primary memory is ROM (Read Only Memory)

- Motherboard: holds CPU, GPU, RAM connectors
- Hard Disk & Solid State Disk (SSD: long state storage

open program → copy data → save data

Software

Set of programs written to perform special tasks or functions on a computer

1. System Software: necessary software to operate a device
2. Application Software: not necessary to operate
   - text processors
   - web browsers
   - media players
   - games

Operating systems

- a software used for interaction between the user and the computer hardware

File Systems

- a structure used by an OS to organise and manage files on a storage device (HD, SSD, USB)

- it defines how data is stored, accessed and organised on the storage device
- ex. your phone is also a file system

Working with files

- Folders: organise files
- HD: a tree of folders and files
- Applications: executable program that used i/o files
- Actions: view, create, or edit (cut, copy, paste, delete) files

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Terminal (=command line)

- a text based method/interface to access information on your computer
- a text input and output environment
- a program that acts as a wrapper and allows to enter commands that the computer processes
- a terminal is faster and more efficient
- task automatisation (the script you created can work overnight)
- sometimes it is the only alternative
- open your command prompt by searching cmd in the search bar

Directories

- you move around your file system content current directory: your working directory
- directories are separated by a slash /
- the topmost directory = a single backslash
- command + any modifiers/arguments

cd/volumes/inaaya/documents/photos

cd/ = move to the top directory

cd. = move to current directory

cd.. = go 1 directory level up in the file tree

pwd = print the complete pathname of the current directory

ls. = listing documents

ls.. = list documents 1 level up

mv <from> <to>: move a file from dir to dir can also re

19th of September (Methodology)

Methods in AI

What is AI?

"AI is the science and engineering of making intelligent machines, especially intelligent computer programs"

~ Alan Turing - godfather of AI

1. Defining the problem
2. Finding knowledge in literature
3. Building AI Systems (this lecture)
    - What do you propose?
    - Why? (motivation)
    - How will your system work (substeps)?

4. Evaluation of AI Systems

- Human vs Machine Intelligence
- Artificial Intelligence and Rationality
  - An “Intelligent Agent” is a device that perceives its environment and takes actions that maximise its chance of successfully achieving its goals.
- Extending the Definition of Intelligence
  - Ability to plan, reason and learn, sense and build some king of perception of knowledge and communicate in natural language

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Types of AI

- Artificial narrow intelligence (ANI), which has a narrow range of abilities
- Artificial general intelligence (AGI), which is on par with humas capabilities
- Artificial superintelligence (ASI), which is more capable than a human

- Categorization of AI
  - capacity to mimic human characteristics
  - the technology they use
  - the real-world application
  - the theory of mind
- Narrow/Weak AI
  - Limited abilities
  - Goal-oriented (perform singular tasks)
  - ex. search, facial recognition, speech recognition/voice assistant, self-driving cars
  - Narrow; small set of constraints and limitations to simulate human behaviour
- General/Strong AI
  - mimics human intelligence and behaviours
  - ability to learn to solve any problem
  - machines think, understand, and act indistinguishably from a human
  - yet to be achieved: challenged by Generative AI
- Super AI (does not exist yet)
  - machines surpass the capacity of human ability
  - currently only in dystopian fiction
  - increasing role in public perception of AI
- Interpretable AI
  - Can we build a Strong AI in a way that we understand?
- Artificial Intelligence: a Socio-Technical Field
  - Artificial Intelligence as a facilitator (if done well)
    - democratising society and empowering humans
    - ex. scaling up production, information provision, creativity etc.
  - Artificial Intelligence as a disruptive technology
    - balance between benefits and risks
  - Artificial Intelligence as (a part of) an Ideology
    - ideology of progress and technological progress
    - ideology of (capitalistic) economic

AI Problems & typical related solutions

- Problem solving: decision making
  - Systematic search through a range of possible actions in order to reach some predefined goal or solution
  - AI into the workflow as a primary processor of data
  - Avoid human cognitive bias
  - Dealing with large data
  - 2 options:
    - Specific-purpose methods
      - tailor-made for a particular problem
      - exploits very specific features of the situation
    - General-purpose methods
      - applicable to a wide variety of problems
  - Heuristics
    - When finding the guaranteed best possible solution using exact methods is too complex
    - a practical method for finding a sub-optimal solution (not optimal, but sufficient for the immediate goal)
      - ex. vehicle routing
- Learning

  Machine Learning (ML) is the scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead.

  1. Supervised learning (classification, regression, there is previous data that you train)
  2. Unsupervised (clustering, you don't have any previous data, you need to find something)
  3. Reinforcement learning (optimizing somethings functionality)
- Reasoning

  Reason is the capacity of consciously making sense of things, establishing and verifying facts, applying logic, and adapting or justifying practices, intuitions, and beliefs based on new or existing information
  - Deductive: produce new explicit statements from implicit ones
  - Inductive: produce explicit statements for unobserved objects
  - Abductive: produce statements that explains known statements
  - Case-based: solve problems based on solutions to similar problems

  Knowledge Representation
  - Making sense of or explaining the world
    - Interpreting perceptions, organising knowledge, conceptional phenomena
  - Useful for:
    - Question answering, decision support, data integration, disambiguation

  Logics
  - Propositions
  - Predicates
  - Objects and instances / Graphs (Description Logics)
  - Modal logics (beliefs, epistemic)
  - Temporal
  - Non-monotonic / Default
  - Fuzzy logics

    - Probabilistic
- Perception
    - The process whereby sensory stimulation is translated into organized experience
    - environment scanned by sensory organs
    - scene is decomposed
        - separate objects
        - spatial relationships
    - ex. autonomous vehicles, trash-picking objects

  Computer Vision
    - object detection, recognition, identification; motion analysis; event and emotion detection
- Social Intelligence
    - Human-Robot interaction
    - AI, Psychology, Human-Computer interaction
    - Machine ethics, moral agency
    - Methods for perceiving humans and motion-planning
    - Cognitive models (theory of mind)
    - ex. health, search & rescue, space exploration
- Natural Language Processing

  Language: a system of signs having meaning by convention (ex. spoken words, traffic signs)
    - Natural language processing
        - understanding the structure of natural language (Syntax)
        - understanding the meaning of natural language (Semantics)
        - understanding the usage of natural language (Pragmatics)
        - language generation
        - machine translation
    - Revolutionised through Transformers and Generative AI in the recent past
- Engineering: Modelling and simulation
    - the use of conceptual models (ex. physical, mathematical, or logical representation of a system, entity, phenomenon, or process) as a basis for simulations to develop data utilized for managerial or technical decision making

  Diagnostics
    - in essence: “this doesn’t work because of this and this and this”
    - Goal: detect faulty behaviour
    - formal model describes the behaviour of a system
    - the diagnosis system simulates the system using the model, and compares the real observations and the predicted ones
    - related to abductive reasoning
- Flowcharts

22nd of September (Guest)

The ethical challenges of artificial intelligence

Data is a piece of information

Ethics are moral principles

3 axes of research

1. Ethics of data
2. Ethics of algorithms
3. Ethics of practices

What is al algorithm

A series of instruction that have a schema, that has to be in a specific sequence that solve a specific problem

Algorithms are vulnerable to people's prejudices, so not entirely neutral

- Regular algorithms follow step by step instruction, are rule based and take some input and some logic to create an output
- Machine learning algorithms do not follow explicit instructions defined by human experts, they're more data-intensive, they learn patterns and they take an input and output to create logic

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Systematicity vs. randomness is what distinguishes machines from humans, where machines are more systematic are humans more random.

- In the context of machine learning, bias refers to systematic errors that skew outcomes in favour of or against certain groups or individuals, not based on merit but on perceived notions, often stemming from the data or the way algorithms are constructed and operated

A low error does not necessarily suggest that the model is reliable and not systematically biased, that depends on where the error occurs. If the errors are consistently impacting a specific group, they are more likely to be biased than if the errors are spread across all groups.

Human biases in data

- unconscious bias from the world that we might reflect in machine learning when using some of the world's data
- reporting bias, selection bias, overgeneralization, out-group homogeneity bias

Human biases in collection and annotation

- unconscious bias in our procedures that we might reflect in our machine learning
- confirmation bias, automation bias

26th of September (Guest)

Conversational AI

Important properties

- Turns
  - A dialogue is a sequence of turns
  - A human needs to know when to stop talking
  - → AI needs to know when to stop talking when talking to a human
- Speech acts/dialogue acts
  - Define what action is being performed by the speaker
  - → conversational AI needs to be aware of the speech acts of the user and itself

- Grounding
  - The speaker and the hearer/reader need to establish what they agree on (= common ground)
  - Ex. acknowledgement can be done with words like uh-huh
  - → conversational AI needs to build and maintain a common ground with the user
- Dialogue structure and subdialogues
  - Ex. adjacency pairs (questions set an expectations for an answer)
  - Dialogues can have side sequences, such as clarification questions, or presequences
  - → AI must understand the structure of the dialogue, including side sequences, questions, assumptions etc.
- Initiative
  - Sometimes a conversation is controlled by one participant, in other cases the initiative is mixed
  - Older AI systems had either system-initiative or user-initiative
  - → balancing initiative is important for AI to have in a meaningful conversation
- Inference
  - The client doesn't seem to answer the user question, but in fact they do
  - The speaker expects the reader/hearer to draw inferences, guided by conversational "maxims":
    - Relevance = speakers don't just utter random words
    - Quantity = speakers say no more and no less than needed
  - → AI must be able to draw inferences from conversations

What does a conversational AI look like?

User speaks → speech recognition → language understanding → dialogue management → language generation → speech synthesis → react to the user

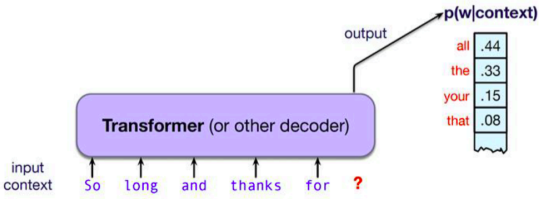
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Large Language Models (LLMs)

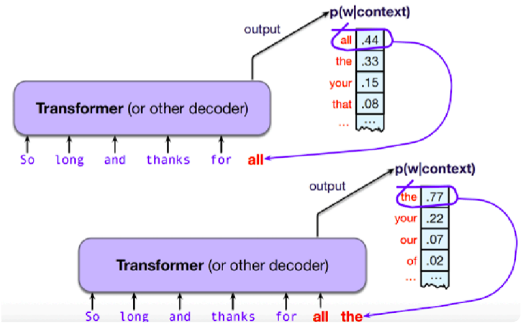
Computational systems that can interact conversationally with people using natural language

LLMs have revolutionized the field of a natural language processing (NLP) and AI

- An LLM is a ("transformer") neural network with:
  - Input: a context of prefix
  - Output: a distribution over possible next words (tokens, which are words or subwords (#ing))



- An LLM that gives a probability distribution over next tokens can generate by repeatedly sampling from the distribution



Decoder Transformers

What people think of LLM:

- GPT, Claude, Llama, DeepSeek, Mistral
- a generative model
- It takes as input a series of tokens and iteratively generates an output token one at a time
- Left to right

Conditional Generation: generating text conditioned on previous text

1. Give the LLM an input piece of text, a prompt
2. Have it generate token by token
    - conditioned on the prompt and the generated tokens
    - steps:
        - computing the probability of the next token wi from the prior context: P(wi|w<i)
        - sampling from that distribution to generate a token

Sentiment Analysis

- The model checks if a statement is positive or negative and acts upon that

Prompt

A text string that a user issues to a language model to get the model to do something useful by conditional generation

- Prompt engineering: the process of finding effective prompts for a task
- In prompting you can have questions, as structured as you want, instructions and demonstrations

*The number of words is smaller or equal than the number of tokens, because tokens can be subwords as well (eating is 1 word but 2 tokens)

Sampling for LLM generation

The internal networks for LLMs generate real-valued scores called logits for each token in the vocabulary

- Softmax is used as a normalization step, it makes sure that the probability factor is shown as a number between 0 and 1

Emphasize high-probability words

+quality: more accurate, coherent, and factual

-diversity: boring, repetitive

Emphasize middle-probability words

+diversity: more creative, diverse

-quality: less factual, incoherent

Greedy sampling

- Take the token with the highest probability
- Extremely predictable (deterministic), the resulting text is generic and repetitive

Random Sampling

- Randomly select a token to generate according to its probability defined by the LM, conditioned on our previous choices, generate it and iterate
- Even though random sampling mostly generate sensible, high probable words, there are many odd, low probability words in the tail of the distribution
- They get picked often enough to generate weird sentences

Temperature Decoding

- Temperature = a knob that allows us to adapt the probabilities, deciding how greedy or how random we want to be with our sampling
  - Low temperature sampling → towards greedy (high probability)
  - High temperature sampling → towards random (more chance on low probability)

Training LLMs

The big idea that underlies all the amazing performance of LLMs:

1. Pretrain a transformer model on enormous amounts of text
   - It happens one time because its expensive and lots of research
   - key idea: text contains enormous amounts of knowledge, pretraining on lots of text with all that knowledge is what gives LLMs their ability to do so much
   - We teach the model to assign a high probability to true word *w*
   - Loss/penalty is high if the model assigns too low probability to *w*
   - Teaching happens through adapting of the weights of the neural network
2. Apply it to new tasks (question answering, sentiment analysis, etc.)

Self-supervised training algorithm

1. Take a corpus of text
2. At each time step t
   1. ask the model to predict the next word
   2. train the model to minimize the error in this prediction

*Self-supervised because it just uses the next word as the label

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Problems with scraping from the web

Copyright: much of the text in these datasets is copyrighted

- not clear if fair use doctrine in US allows for this use
- this remains an open legal question across the world

Data consent

- website owners can indicate they don't want their site crawled

Privacy

- Websites can contain private IP addresses and phone numbers

Skew

- Training data is disproportionately generated by authors from the US which probably skews resulting topics and opinions

Challenges with data filtering

Quality is subjective

- many LLMs attempt to match Wikipedia, books, particular websites
- need to remove boilerplate, adult content
- deduplication at many levels (URLs, documents, even lines)

Safety is also subjective

- Toxicity detection is important, although that has mixed results
- Can mistakenly flag data written in dialects like African American English

29th of September (Methodology)

Empirical Evaluations

The path towards a Tehnological Intervention

Whenever you have a problem in real life and you use technology to solve it

Steps to make a valid system

- Designing the system - define system requirement and target groups
- Implementing the system - design system architecture and inference engine, code, programming
- Testing the system - evaluate verification and fine-tune parameters
- Using the system - validation + user evaluations, does the system help its users achieve their goal

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Verification vs Validation

- Verification is the process of checking whether the software system meets the specified requirements of the users → testing the system's Correctness
  - Is it able to provide users with personalized plans targeting individual users' need?
  - For building the system
  - Internal validation of verification
- Validation is the process of checking whether the software system meets the actual requirements of the users → testing the system's Effectiveness
  - Does the system actually work?
  - For using the system
  - External validation

External validation is best achieved by means of a *Randomized Controlled Trial*

- You randomly divide the users into a treatment group (exposed to the technological intervention) and a control group (not exposed to the technological intervention), then you follow-up on both of them and compare the results
- You have to run a statistical test to check if the results are significant
  - the primary output of statistical tests is the p-value (probability value), it indicates the probability of observing the difference if no difference exists
  - the easiest type of statistical test is a t-test comparing the performances of two groups of users

Research protocol

- Research question - the objective of a study or a problem to be solved through research
- Hypothesis - a proposed explanation for a phenomenon
- Experimental and Control group - respectively, the group of participants who are exposed to the technological intervention and the group who are not exposed to the technological intervention
- Independent variable - the variable that is changed or controlled in a scientific experiment to test the effects on the dependent variable (app for drinking behavior)
- Dependent variable - the variable being tested and measured in a scientific experiment (the drinking behavior itself)
- Subjects/participants - the people taking part in the experiment/intervention
- Possible confounding variables - extra independent variables that are having a hidden effect on your dependent variable

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How can you test a research question?

- Enquête, ask the users if it worked or not
- Using a sensor, so you can track it yourself
- Use two completely different things and compare the results

User Experience Evaluation

- User Experience (Usability) - do you think it works? Is it easy to use? Will you keep using it? etc.
- Questionnaires (based on likert-scales) - used to evaluate user experiences
  - Acceptability - refers to the factors that affect the users' willingness to use the application
  - Usability - refers to the design factors that affect the user experience of operating the application's device and navigating the application for its intended purpose
  - Feasibility - refers to whether the intervention can actually be carried out or not

3rd of October (Guest)

Artificial Intelligence and Human Behavior

- Human behavior is foundational for the health of our society
- AI technology is omnipresent and influencing our behaviors
  - however, not always in a responsible and ethical way
- Many societal challenges require different behaviors
  - Population health
  - Climate change
  - Democracy and justice
- AI can contribute to a better understanding of human behavior and supper behavior change

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Research Framework

1. Understanding Human Behavior
   - interpret, simulate, reason
   - knowledge driven, data driven
2. Supporting Behavior Change
   - supportive approach

- non manipulative
- avoiding tech-dependence

Understanding Behavior

"Reading a person's mind"

- We use a computer model to describe what happens in a human's mind
- Such models help us to
  - understand decision making
  - make predictions
  - decide on the support
- "theory of mind" - Psychology

Multidisciplinary Research

Understanding, and influencing human behavior is multidisciplinary

- Psychology
  - what is human behavior?
  - which techniques can be used to change behavior?
  - how to do user studies?
- Communication Science
  - persuasive communication
- Economics
  - incentives
  - persuasive technology
- Philosophy
  - human autonomy
  - ethics and values

Many relevant computational techniques

- Predictive modelling (data driven or knowledge driven)
- Simulation
- Virtual Reality
- Machine learning / pattern recognition
- Chatbots / LLMs
- Data from wearables
- Social media analysis

Ten Challenges

1. Finding the thin line between motivation and manipulation
   - How do we encourage volitional decision making, instead of reducing their autonomy?
2. How to support persons to grow?
   - How do we make sure we do not make people dependent on technology?
   - Can we help people learning new things instead of becoming less skilled?
3. How to develop systems that looks at the needs of the whole person?

    - How do we prevent that we optimize on one behavior at the cost of another?
4. How to prevent overreach?
    - How to set objectives that are stimulating but not too far away?
5. How to focus on user needs and how to best support them
    - How do we make sure solutions are really supporting a persons objectives?
    - How to prevent that technology is seen as the only and best solution?
6. How to integrate BC technology within the wider (social) environment?
    - How to take the social and physical context of people into account?
7. How to anticipate long-term effects?
    - Change should be sustainable and the support should evolve over time
    - Societal consequences of individual behavior changes should be taken into account
8. How to evaluate behavior change?
    - How to select, work-out and implement mixed-method evaluations for BCSSs (Behavior Changing Support Systems) during their lifecycle in a coherent way?
9. How to make explicit what values underlie the different tools/techniques, and how to align that with user values?
    - How to align AI goals with human goals in a way that the human goals are supported?
10. Responsible development
    - How to ensure data privacy and that sensitive contextual user information is not exploited?
    - How to address fairness?

Theoretical Background

- Behavior is complex, interplay of individual and environmental factors
- Behavior cannot be changes directly; instead, interventions work on parts of the human psyche, or psychological constructs
    - → determinants of behavioral change
- Interventions usually target
    - multiple determinants
    - with multiple components
    - ???

(sub)determinants we should focus on

- Selecting determinants based on quantity and quality evidence for results found
    - Quantity = how many studies provide evidence for this relationship
    - Quality = what is the methodological quality of these studies?
- Selecting determinants based on relevance and changeability
    - Relevance = how strong is the relationship between determinant and behavior
    - Changeability = how much influence do you have on changing determinant

(sub)determinants that play an important role according to psychological theory

- Theory of Planned Behavior
    - Attitude + perceived norm + perceived behavioral control → intention (+ skills) behavior
- Social Cognitive Theory
    - Knowledge, expectations, social norms, skills, attitudes → determinants of behavior
- Fear-Avoidance Model

- Pain catastrophizing ⬆ → pain anxiety + kinesiophobia (fear of moving) ⬆ → avoidance of behavior ⬆ = physical activity ⬇ → depressive symptoms ⬆

Emotion contagion - if you're aggressive people around you will most likely become aggressive too, while people further away won't

6th of October (Methodology)

Discrete Mathematics

Naive Set Theory

A set is a structure representing an *unordered* collection of zero or more *different* objects

- set theory deals with operations between, relations among, and statements about sets
- Notation: { }

We use variables, ex. S, T, U etc.

- We denote a set S in curly brackets with all the variables inside: S = {a, b, c}

Set builder notation: for any proposition P(x) (statement), {x/P(x)} is the set of all x such that P(x)

- ex. {x | x is an integer where x > 0 and x <5} = {1, 2, 3, 4}

Sets are unordered so {a, b, c} = {b, c, a} = {a, c, b}

Sets are distinct (unique) so {a, b, c} = {a, a, b, b, b, c}

Sets can be never ending (infinite)

Venn Diagrams = model of sets

The universe (of discourse) *U* is the set of all things

x ∈ S means "x is an element or member of set S"

- "x is in S"

Set A is equal to set B *iff* they have the same exact members

- x ∉ S *iff* x is not in S

∅, null or empty set, is the unique set that contains no elements

The symbol for subset is ⊂ (proper subset) or ⊆ (subset or equal to)

- S ⊆ T, every element of S is also an element of T
- S ⊂ T, every element of S is in T, but not the other way around

The symbol for superset is ⊃ (proper superset) or ⊇ (superset or equal to)

- S ⊇ T, every element in T is in S, but not necessarily vice versa

The elements in a set may be sets themselves

|S| means "the cardinality of S, or the measure of how many elements S has"

- |{1, 2, 3}| = 3
- |∅| = 0
- |{1}, {1{2, 3}}| = 2

The power set P(S) of a set S is the set of all subsets of S

- P(S) = {x | x ⊆ S}
- Sometimes P(S) is written as 2 to the power of S

A ∪ B means “A union B”, or the set containing all elements that are either in A or in B

- A = {1, 2, 3} and B = {4, 5, 6} means A ∪ B = {1, 2, 3, 4, 5, 6}

A ∩ B means “A intersection B”, or the set containing all elements that are both in A and B

- A = {1, 2, 3} and B = {3, 4, 5} means A ∩ B = {3}

A ∩ B = ∅ means “A and B are disjoint”, or their intersection is empty

- A = {1, 2, 3} and B = {4, 5, 6} means A ∩ B = ∅

A - B means “the difference of A and B”, or the set of all elements in A but not in B

- A - B = {x | x ∈ A and x ∉ B}
- Also called “the complement of B with respect to A”

A’ is “the complement of A”, or everything in the universe that is not A **...?**

Identity
Domination
Idempotent
Double complement
Commutative
Associative
De Morgan

Relations

A list is a structure representing an *ordered* collection of zero or more objects which may include *duplicates*

- Notation: [ ], ( ), ⟨ ⟩
- Elements in a list have positions
- Finite lists have a length

Cartesian product of two sets, a set of pairs (lists of length x 2)

- A x B = { ⟨a, b⟩ : a ∈ A and b ∈ B}
- A x A = A to the power of 2

N-tuples are sequences of lengths n, whose order matters

- (a) is a singles
- (a, b) is a pair
- (a, b, c) is a triple

Relations are sets of n-tuples

A relation in the set A is a binary relation of type A to the power of 2

A binary relation is a relation of type A x B or A to the power of 2

- FollowsCourse = {⟨x, y⟩ : x follows y} ⊆ Students x Courses
  - Alternative infix notation: x FollowsCourse y instead of ⟨x, y⟩ ∈ FollowsCourse

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Graphs

A graph G is generalisation of the simple concept of a set of dots, links, edges, or arcs

- G = (V, E), where V is the set or vertices (singular: vertex), and E is the set of paired vertices or edges

Undirected graphs

- A non-empty set of vertices V, and a set of unordered, distinct edges E
  - G(V, E) V = {u, v, w} and E = {{u, v}, {v, w}, {u, w}}

Directed graphs

- A non-empty set of vertices V, and a set of directed edges E, consisting of ordered pairs
  - G(V, E) V = {u, v, w} and E = {(u, v), (v, w), (u, w)}

Definition: A graph is Eulerian if it is both connected and has a closed trail (a walk with no repeated edges) containing all edges of the graph

Theorem: A graph is Eulerian if every node in the graph has an even degree

The adjacency matrix of a graph G is the matrix A defined as adjacency between nodes u and v

- if the set of edges E includes (u, v), then A[u, v] has a value of 1
- if there is no direct path to u to v, then A[u, v] has a value of 0

Trees

A tree is a connected, undirected graph with no simple circuits

- A circuit (cycle) is a path of length ≥ 1 that begins and ends at the same vertex
- Ex. tournaments and family trees

Once a tree has a designed root node, it is possible to assign a direction to each edge

- Root node is the one at the top
- Internal nodes are the nodes in between
- Parent relations, one above the other
- Sibling relations, the same order or the same parent
- Leaf node is the last node

A rooted tree is an n-ary tree if every internal vertex has no more than n children

- A tree is full n-ary if every internal vertex has exactly n children
- An n-ary tree with n=2 is called a binary tree

The level or depth of a vertex v in a rooted tree is the length of the unique path from the root to this vertex

The height (also called depths) of a rooted tree is the maximum of the levels of vertices

10th of October

Neurosymbolic AI

Data-driven methods to learn

- Supervised learning
  - Providing labeled data to check if the answer is right by telling the correct answer
- Unsupervised learning
  - Lets the machine figure out the right answer without handing it to them
- Reinforcement learning
  - Learning with rewards and punishments (not necessarily from humans, can also be from the algorithm itself)
  - Often needs unreasonably many trial-and-errors

- Natural way of learning

Knowledge-driven methods to learn

13th of October

Evolving embodied intelligence

Digital AI - Thinking machines

- machine translation
- chatbots
- image analysis

Embodied AI - Acting machines

- robot receptionist
- warehouse robots
- self-driving cars

What is a robot?

- Technically: a machine that can sense, compute and act
- Conceptually